

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2021-2022
FULL MARKS: 200

Math 4241: Integral Calculus and Differential Equations

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

- ✓ 1. a) Consider the function: $f(x) = 3x + 2x^2$. 12
 Find a formula for the Riemann sum obtained by dividing the interval $[0, 1]$ into n equal subintervals and using the right-hand endpoint for each c_k for the given function. Now, take a limit of these sums as $n \rightarrow \infty$ to calculate the area under the curve over $[0, 1]$. (CO2)
(PO1)
- b) State and prove the 2nd fundamental theorem of calculus. 11
(CO1)
(PO1)
- c) Pick a point (x, y) at random in the interval $0 < y < 1 - x^2$. What is the probability that $x > \frac{1}{2}$? 10
(CO2)
(PO1)
- ✓ 2. a) "A potential function is called a conserved quantity"-Explain with appropriate example. 5
(CO1)
(PO1)
- b) Find the area bounded by the parabola $y^2 = 4x$ and the line $2x - y = 4$. 12
(CO2)
(PO1)
- c) Calculate each of the following integrals: 8 × 2
(CO2)
(PO1)
- i. $\int \frac{\sqrt{x^2 - a^2}}{x} dx$
- ii. $\int \frac{6x^2 + 14x - 20}{x^3 - 4x} dx$
- ✓ 3. a) What do you understand by order and degree of any differential equation? 5
(CO1)
(PO1)
- b) Compute the volume of the solid generated by revolving the region bounded by $y = x$ and $y = x^2$ about each coordinate axis using both shell method and washer method. 16
(CO2)
(PO1)
- c) Find the length of the curve $x = \frac{y^2}{3} - y^{\frac{1}{2}}$ from $y = 1$ to $y = 9$. 12
(CO2)
(PO1)

✓ 4. a) What is the difference between ordinary differential equations and partial differential equations?

b) Find every solution to the following Bernoulli equation:

$$y' = \frac{3}{t}y + t^4y^{1/3}$$

c) Consider the following differential equation:

$$\frac{d^2y}{dx^2} - 5\frac{dx}{dy} + 6y = 0$$

- i. Determine two linearly independent solutions to this equation on the interval $-\infty < x < \infty$. How do you prove their independence?
- ii. Provide a general solution for this differential equation (if exists).
- iii. Find the solution that satisfies the conditions: $y(0) = 2, y'(0) = 3$.

✓ 5. a) Find the explicit solution to the following Initial Value Problem (IVP):

$$(t^2 + 2ty)y' = y^2, y(1) = 1$$

b) Find the solution to the IVP below in explicit form:

$$y' = \frac{2-t}{1+y}, y(0) = 1$$

c) Consider the equation:

$$(6x^5 - xy) + (-x^2 + xy^2)y' = 0 ; \text{ with initial condition } y(0) = 1.$$

- i. Find an integrating factor μ that converts the above equation into an exact equation.
- ii. Find an implicit expression for the solution to y of the IVP.

✓ 6. a) A vessel with liquid at 18°C is placed in a cooler held at 3°C , and after 3 minutes the temperature drops to 13°C .

- i. Find the differential equation satisfied by the temperature T of a liquid in the cooler at time $t = 0$.
- ii. Find the function temperature of the liquid once it is put in the cooler.
- iii. Find the liquid cooling constant.

b) Solve the following equation using Lagrange's method:

$$p + 3q = 5z + \tan(y - 3x)$$

c) Solve the following equation using Charpit's method:

$$(p^2 + q^2)y = qz$$

12
(CO2)
(PO1)
12
(CO2)
(PO1)
5 × 2
(CO2)
(PO1)

4 × 3
(CO2)
(PO1)

10
(CO2)
(PO1)
12
(CO2)
(PO1)